an NBBO;[1644] (2) the best bids and best offers from each SRO;[1645] and (3) information on trades such as prices and sizes. The SIP data also includes certain regulatory data, such as information required by the LULD Plan,[1646] information relating to regulatory halts and MWCBs,[1647] information regarding short sale circuit breakers,[1648] and other data, such as data relating to retail liquidity programs, market and settlement conditions, the financial condition of the issuer, OTC equities, last sale prices for corporate bonds, and information about indices.[1649]

The exchanges separately sell their individual market data directly to market participants via proprietary data feeds. For example, the exchanges have developed proprietary DOB products that provide greater content (e.g., odd-lot quotations, orders at prices above and below the best prices, and information about orders participating in auctions, including auction order imbalances) at lower latencies,[1650] relative to the exclusive SIPs, for certain segments of the data

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1644 The national best bid and offer are constructed from the best bid and offer prices across all exchanges in which the quoted size is at least one round lot. See Proposing Release, 85 FR at Section III.C.1.

1645 The best bids and offers on an exchange are determined by the best prices in which the quoted size is at least one round lot. Some exchanges aggregate odd-lot orders at better prices into round lots and report such aggregated orders as their best bid or offer at the least aggressive price of the aggregated orders. Typically, the best bids and offers on each exchange are protected quotes under NMS Rule 611 and cannot be traded-through. See Proposing Release, 85 FR at Section III.C.1(a).

1646 See Proposing Release, 85 FR at n.38.

1647 See id. at n.39.

1648 See id. at n.40.

1649 See id. at n.41.

1650 See, e.g., Proposing Release, 85 FR at n.19 (for Nasdaq Global Data Products, Real-Time - NYSE Proprietary Market Data, and Cboe Equities Offerings, all describing low-latency DOB data products). Commenters agreed with this description of the market. See, e.g., DOJ Letter at 2; Committee on Capital Markets Regulation Letter at 2.

market, such as automated trading systems. They have also developed proprietary TOB products that provide data that is generally limited to the highest bid and lowest offer and last sale price information and are typically priced lower than the SIP data for another segment of the data market that is less sensitive to latency (e.g., retail or non-professional investors and wealth managers that access market data visually).[1651] Proprietary data feeds are available as part of exchanges' standard offerings. Most exchanges offer for sale as part of their proprietary DOB products the complete set of orders at prices above and below the best prices (e.g., depth of book data), complete odd-lot quotation information, and information about orders participating in auctions, including auction order imbalances (for listing exchanges).[1652]

1651 Examples of such proprietary TOB products include NYSE BBO, Nasdaq Basic, and Cboe One Feed. See Proposing Release, 85 FR at n.19. NYSE BBO provides TOB data. Nasdaq Basic and Cboe One's Summary Feed provide TOB and last sale information. Nasdaq Basic also provides Nasdaq Opening and Closing Prices and other information, including Emergency Market Condition event messages, System Status, and trading halt information. Cboe One also offers a Premium Feed that includes DOB data. Each of these products is sold separately by the relevant exchange group. See Letter from Matthew J. Billings, Managing Director, Market Data Strategy, TD Ameritrade, (Oct. 24, 2018) ("TD Ameritrade Letter 2018"), available at https://www.sec.gov/comments/4-729/4729-4560068176205.pdf at 5–8 (stating that the lower cost of exchange TOB products, coupled with costs associated with the process to differentiate between retail professionals and non-professionals imposed by the Equity Data Plans, and associated audit risk, favors retail broker-dealer use of exchange TOB products).

1652 See Proposing Release, 85 FR at n.335. IEX and MEMX make proprietary data available but do not charge for it. See, e.g., IEX, Market Data, available at https://iextrading.com/trading/market-data/ (last accessed Jan. 8, 2020); MEMX Fee Schedule, available at https://info.memxtrading.com/fee-schedule/ (last accessed Nov. 18, 2020). See also Ramsay Letter II. Long Term Stock Exchange does not offer a proprietary data feed, but makes information on the order book available on its website. See, e.g., LTSE Connectivity Guide, available at https://assets.ctfassets.net/cchj2z2dcfyd/1Jp5V4TWZXzhl8QmBWD3Ed/2f926fa4c55f6f489cceb8b77fe8e685/LTSE_Connectivity_Guide.pdf (last accessed Nov. 18, 2020).

One notable gap between SIP data and proprietary DOB data is that SIP data does not include complete odd-lot quotation information even though odd-lots represent a large share of all trades in the U.S. stock market and can represent economically significant trading opportunities at prices that are better than the prices of displayed and disseminated round lots.[1653] While several exchanges aggregate odd-lot orders into round lots and report such aggregated orders as quotation information to the exclusive SIPs,[1654] market participants must purchase proprietary data feeds, available from the exchanges, to see the odd-lot quotations that are priced at or better than the best bid or offer.[1655]

Odd-lot transactions make up a significant proportion of transaction volume in NMS stocks, including ETPs, and a significant proportion of odd-lot trades occur at prices better than the prevailing NBBO, especially in higher priced stocks. In May 2020, approximately 45% of all trades executed on exchanges and approximately 10% of all volume executed on exchange in corporate stocks and ETFs were odd-lot sized and that 40% of those transactions (representing

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1653 See Alexander Osipovich, *NYSE Aims to Speed Up Trading with Core Tech Upgrade*, WALL STREET JOURNAL (Aug. 5, 2019), available at https://www.wsj.com/articles/nyse-aims-to-speed-up-trading-with-core-tech-upgrade-11565002800 (Retrieved from Factiva database). Commenters agreed that odd-lot quotes make up a significant portion of trading interest, especially in higher priced stocks. See, e.g., Cboe Letter at 6; Healthy Markets Letter I at 3; Clearpool Letter at 11–12; IntelligentCross Letter at 3; IEX Letter at 3–4; ICI Letter at 7.

1654 See Proposing Release, 85 FR at Section III.C.1(a). Exchange rules specify how the aggregation process works in different terms and with different levels of specificity, but many exchanges aggregate odd-lots across multiple prices and provide them to the exclusive SIPs at the least aggressive price if the combined odd-lot interest is equal to or greater than a round lot. See Proposing Release, 85 FR at nn.157, 158, 789.

1655 See Proposing Release, 85 FR at n.163. Commenters agreed that the absence of odd-lot quote information reduces the usefulness of the SIP. See, e.g., BlackRock Letter at 3;

approximately 35% of all odd-lot volume) occurred at a price better than the NBBO.[1656] Additionally, a significant portion of quotation and trading activity occurs in odd-lots, particularly for frequently traded, high-priced tickers, and as stock prices rise, the difference in spreads calculated using the different feeds also rises, indicating that odd-lots are more likely to set the best quote as stock prices rise.[1657]

A number of commenters also submitted analyses examining the occurrence of odd-lot trades. Commenter analyses generally observed that odd-lot trades occur frequently in higher priced stocks and that their frequency has increased over time, along with an increase in the

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1656 See supra note 241. Similar staff analysis in the Proposing Release examining a different time period also showed that odd-lot trades account for a significant proportion of transactions. See Proposing Release, 85 FR at 16813.

1657 See supra note 240. The staff analysis found that for the 500 top tickers by dollar volume, odd-lot quotes represented a significant price improvement over the exclusive SIP quotes. This analysis further found that as the price of the stock increased, the duration-weighted amount by which the odd-lot quote improved on the SIP quote increased as well. Similar staff analysis in the Proposing Release examining a different time period found similar results. See Proposing Release, 85 FR at Section III.C.1(b). Analysis by one commenter also observed that there are frequently odd-lot limit orders priced better than the NBBO and that this is more common in higher priced stocks. See JP Morgan Memo to File at 2.

average stock price.[1658] Commenter analysis also observed that odd-lot limit orders occur frequently in higher priced stocks.[1659]

One commenter stated that odd-lot trade frequency is not a valid proxy for passive order interest because trade size is often determined by the liquidity-taking order and is often a result of algorithmic "pinging."[1660] This commenter conducted an analysis and concluded that it is small liquidity-taking orders that are driving the increase in odd-lot trades.[1661] This commenter did not observe an increase in the size of passive retail investor orders but did find a decrease in their execution size. The Commission acknowledges that algorithmic "pinging" could account for a portion of the odd-lot trading volume that occurs but also believes that odd-lot-sized limit orders can represent a significant source of liquidity, especially in higher priced stocks. This commenter's analysis was limited to the orders of retail investors, while the staff analysis discussed above and the analyses submitted by other commenters, which observed that odd-lot limit orders are a significant source of liquidity (especially in high-priced stocks), contained the

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1658 Analysis from a number of commenters observed that odd-lot trades are more prevalent in high priced stocks. See, e.g., BestEx Research Letter at 6; Capital Group Letter at 3; Nasdaq Letter III at 11; RBC Letter at 5. Additional commenter analysis observed that the percentage of odd-lot trades has increased over time, especially in high priced stocks. See, e.g., BestEx Research Letter at 6; Healthy Markets Letter I at 11–12. Analysis from one commenter also observed that the frequency of odd-lot trades also increased off-exchange. See Healthy Markets Letter I at 11–12. Commenters also observed that the average stock price has increased over time. See, e.g., Cboe Letter at 6 ("The average price of a stock included in the S&P 500 Index was $44.86 at the end of 2005, compared to $140.47 at the end of 2019"); Virtu Letter at 3; Citadel Letter at 3. Other commenters also agreed that odd-lot trading has increased over time, especially in high priced stocks. See, e.g., Fidelity Letter at 4; ACS Execution Services Letter at 2; Angel Letter at 13.

1659 See, e.g., Nasdaq Letter IV at 16; Schwab Letter at 4; Citadel Letter at 3.

1660 See TD Ameritrade Letter at 6–7.

1661 See id.

orders of other types of traders.[1662] Additionally, analyses from other commenters also observed that a significant portion of retail limit orders are smaller than 100 shares, and that this is more common in higher priced stocks.[1663]

Information on odd-lot quotes can help with the optimal placement and routing of orders across markets.[1664] Odd-lot quotation data can help market participants improve trading strategies and lower execution costs by allowing them to take advantage of odd-lot quotes that are available at prices better than the NBBO, possibly on a different exchange than where the NBBO is located. Odd-lot quotation data can also help market participants place limit orders at prices at or inside the NBBO. SIP data is unable to differentiate between individual round lot quotes and odd-lot quotes that were aggregated by the exchanges to be a round lot quote.

Another gap between SIP data and proprietary DOB data is that SIP data currently lacks quotation information in NMS stocks beyond the top of book[1665] even though the decimalization

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1662 See supra note 1657. See also, e.g., Nasdaq Letter IV at 16; JP Morgan Memo to File at 2.

1663 See, e.g., Nasdaq Letter IV at 16; Schwab Letter at 4 ("In the first quarter of 2020, a total of 1.87 million, or 23 percent, of Schwab customers' limit orders for stocks priced higher than $100 are for fewer than 100 shares"); Citadel Letter at 3. One commenter also stated that retail investors tend to trade in lots smaller than 100 shares. See Schwab Letter at 4.

1664 Commenters agreed that odd-lot information has become important for trading decisions. See, e.g., ACS Execution Services Letter at 2; Clearpool Letter at 11–12; IntelligentCross Letter at 4. A panelist at the Roundtable also stated that odd-lot quotation data is needed to make effective decisions in trading applications and to fill client orders effectively. See Proposing Release, 85 FR at n.173 and accompanying text.

1665 See Proposing Release, 85 FR at Section III.C.2.

of securities pricing in 2001 led to a dispersion of quoted volume away from the top of book.[1666] Consequently, the NBBO shown in SIP data became less informative and some market participants have come to view depth of book data as necessary to their efforts to trade competitively and to provide best execution to customer orders.[1667] Market participants interested in such depth of book data must rely upon the proprietary DOB products offered by the exchanges that include varying degrees of depth data.[1668]

Staff analyzed depth of book quotations for corporate stocks using data from the week of May 4, 2020 and found that there is a substantial amount of quotation volume at several levels below the best bid.[1669] The analysis also found that during active parts of the trading day, there is quotation interest at every $0.01 increment at least ten levels out for the most liquid stocks; for the least liquid stocks, there is a large gap between the best bid and the next highest bid and large gaps are generally also present between the next several bid levels. Additionally, the analysis found a significant percentage of the total notional value of all depth of book quotations for both liquid and illiquid stocks falls within the first five price levels.

1666 Commenters agreed that decimalization led to a decline in top of book liquidity. See, e.g., Schwab Letter at 3; ACS Execution Services Letter at 2.

1667 See Proposing Release, 85 FR at Section III.C.2(d).

1668 See Proposing Release, 85 FR at n.270. Commenters stated that top of book information is insufficient and market participants pay for proprietary feeds to access depth of book information. See, e.g., ICI Letter at 9.

1669 See supra note 387. Similar staff analysis in the Proposing Release examining a different time period found similar results. See Proposing Release, 85 FR at Section III.C.2(d). Commenters also referenced analysis that observed there was significant liquidity beyond the top of book. See, e.g., RBC Comment Letter at 4–5 (referencing an analysis RBC had previously submitted to the Commission); IEX Letter at 5 (referencing an academic study by Tolga Cenesizoglu and Gunnar Grass, *Bid- and ask-side liquidity in the NYSE limit order book*, 38 J. FIN. MKTS. 14 (2018)).

The Commission recognizes that market participants have diverse market data needs. Depth of book data can assist SORs and electronic trading systems with the optimal placement of orders across markets.[1670] Specifically, depth of book data can help market participants improve trading strategies and lower execution costs by placing liquidity taking orders that are larger than the displayed best bid or best offer and achieve queue priority for liquidity providing orders that post at prices away from the best bid or offer.[1671] At the same time, the depth of book data may be less valuable to a certain segment of market participants (e.g., some retail or non-professional customers).[1672] For example, a relatively small portion of marketable orders execute at prices outside the NBBO indicating that some market participants submitting marketable orders do not find "walking the book" useful.[1673]

1670 Several commenters agreed that depth of book information is useful in routing and placing orders effectively. See, e.g., Clearpool Letter at 14; IntelligentCross Letter at 4; IEX Letter at 5; Schwab Letter at 3; Angel Letter at 9.

1671 See infra Section V.C.1(c)(ii). Commenters agreed that depth of book information helps with placing market orders and accessing liquidity beyond the top of book. See, e.g., ICI Letter at 8–9; Schwab Letter at 3; ACS Execution Services Letter at 4–5; IEX Letter at 5.

1672 One commenter believes that depth of book information would be valuable for retail investors in less liquid stocks and for placing limit orders. See Angel Letter at 1–9.

1673 That is, a marketable order so large that it executes against all the volume at the top of the book and then executes against orders behind the top of the book. See Craig W. Holden and Stacey Jacobsen, *Liquidity Measurement Problems in Fast Competitive Markets*, 69 J. FIN. 1760, at Table I (2014) (showing that 3.3% of orders clear outside the NBBO). This does not necessarily mean that limit orders outside the NBBO are irrelevant. There are limitations to using the observation of trades at prices outside the NBBO at the time of trade execution as an indicator for orders that executed at prices outside of the NBBO at the time of trade order (specifically, these events are not necessarily the same thing). Additionally, instead of submitting a large marketable order that "walks the book", market participants may split a larger marketable order into smaller child orders, with some smaller orders executing against liquidity providing orders at the top of the book and others later executing against liquidity providing orders that were behind the top of book when the first child orders executed. See infra Section V.B.3(e).

Another gap between SIP data and proprietary DOB data is that SIP data includes only limited auction-related information. Auctions are important liquidity events, accounting for approximately 7% of daily equity trading volume.[1674] Closing auctions generate prices that are used for a variety of market purposes, including setting benchmark prices for index rebalances and for determining NAV prices for mutual funds and ETFs.[1675] Auctions are important for the implementation of passive investment strategies. For example, one commenter stated that mutual funds and ETFs that utilize passive index-tracking strategies actively participate in closing auctions.[1676] One commenter stated that reopening auctions play an important role in connection with security-specific or market-wide events, such as a limit up-limit down or other regulatory halt.[1677] Auction imbalance information and indicative prices can help facilitate order placement in auctions and predict price movements.[1678]

1674 See supra note 466 for staff analysis; see also Proposing Release, 85 FR at Section III.C.3(c) and n.348. Commenters agreed that an increase in the portion of total trading volume executed in opening and closing auctions makes them important liquidity events. See, e.g., Cboe Letter at 21; Clearpool Letter at 15; MEMX Letter at 5–6; IEX Letter at 6; Fidelity Letter at 5; Schwab Letter at 5; ACS Execution Services Letter at 2; Angel Letter at 8; Data Boiler Letter I at 31. A number of commenters attributed the growth in auction volume to the increase in passive investing. See, e.g., Schwab Letter at 5 ("The growth of passive investing and exchange-traded funds (ETF) has contributed to the growth in auctions relative to other trading."); SIFMA Letter at 5. See also Proposing Release, 85 FR at 16735.

1675 Commenters agreed with this assessment. See, e.g., Clearpool Letter at 15; MEMX Letter at 5–6.

1676 See ICI Letter at 9.

1677 See Clearpool Letter at 15.

1678 See, e.g., ICI Letter at 9 ("Auction information, which includes imbalance levels between buy and sell orders, allows funds to decide whether to participate, and if so, to determine direction, order size and timing."); BlackRock Letter at 2 ("Auction information

Today, some NYSE auction data, such as pre-opening indicators,[1679] are disseminated through the CTA/CQ SIP, and no auction information generated by the other primary listing exchanges is distributed through the exclusive SIPs, except very limited LULD information related to auction collar messages.[1680] Thus while the exchanges' proprietary data includes detailed information on several aspects of their auctions, only a small subset of the auction-related information is included in SIP data.[1681]

While all listing exchanges make auction information available to market participants through proprietary data feeds, only some exchanges offer this information through specialized feeds for a lower price than full DOB products. For instance, NYSE Order Imbalances is an example of such a proprietary auction data product offered by NYSE,[1682] while Nasdaq does not offer such a specialized product.[1683]

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telegraphs the direction and magnitude of price moves at the end of the day."); MEMX Letter at 6.

1679 See NYSE Rule 15.

1680 See Proposing Release, 85 FR at n.333; see also UTP Plan, UTP Participant Input Specification (Dec. 3, 2019), available at http://www.utpplan.com/DOC/UtpBinaryInputSpec.pdf.

1681 See, e.g., NYSE, TAQ NYSE Order Imbalance – Quick Reference Card, available at https://www.nyse.com/publicdocs/nyse/data/TAQ_NYSE_Order_Imbalance_QRC.pdf (last accessed Jan. 8, 2020).

1682 See NYSE, Real-Time Data Imbalances, available at https://www.nyse.com/market-data/real-time/imbalances (last accessed Jan. 8, 2020) (describing the NYSE Order Imbalances product).

1683 The Nasdaq Net Order Imbalance Indicator is a feature of Nasdaq's BookViewer proprietary data feed product rather than a stand-alone product. See Nasdaq, Net Order Imbalance Indicator, available at https://data.nasdaq.com/NOII.aspx (last accessed Jan. 8, 2020).